Modeling Railway Systems in Answer Set Programming

Ryan Kepler J. Murphy

University of Potsdam



Outline

- Introduction
- Background
- Research goals
- Current status
- Preliminary results
- Open topics

INTRODUCTION

Motivation

Why model railway systems?

- Personal motivation
- History of research

Why Answer Set Programming?

- Understanding ASP
- Benefit of ASP

BACKGROUND

Flatland

What is Flatland?

Grid-based virtual railway environment.



- Developed for global railway scheduling competition
- The objective is to navigate trains to their goals without collisions and before their deadlines

RESEARCH GOALS

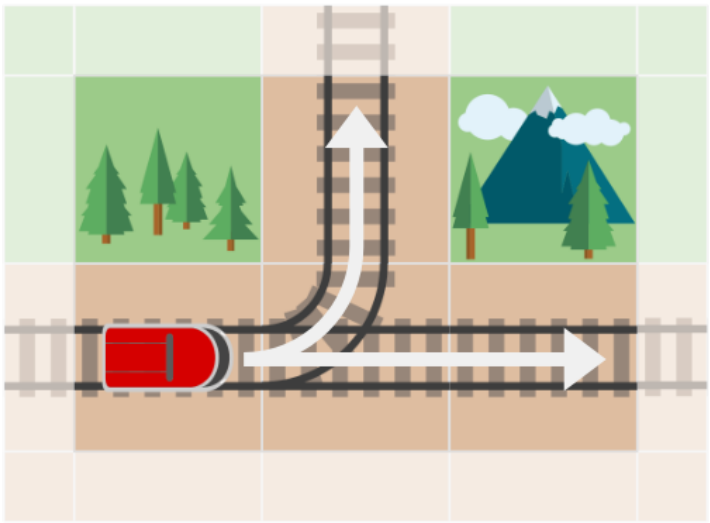
Three phases

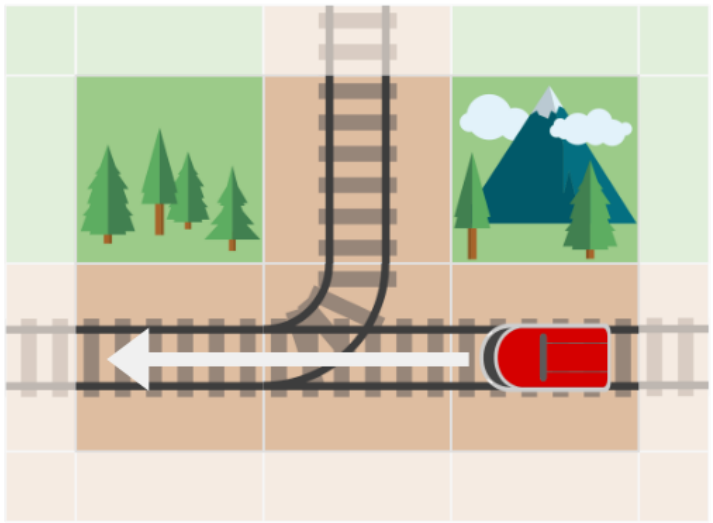
1. Formalization of railway systems
2. Solutions to the Flatland problem
3. Cross-domain transfer

Three phases

1. Formalization of railway systems

- Restricted movement
- Importance of direction





Three phases

2. Solutions to the Flatland problem

VSP

In the vehicle scheduling problem (VSP), vehicles make trips over a transportation network while avoiding collisions and meeting deadlines.[1]

VRSP

The vehicle rescheduling problem (VRSP) is the problem of finding a new schedule that not only minimizes the total traveling costs but also minimizes the costs of deviating from the original schedule.[2]

1. (Gharibi, Waslander, and Boutaba; 2020)
2. (Spliet, Gabor, and Dekker; 2014)

Three phases

3. Cross-domain transfer

How do our findings apply to other domains and vice versa?

CURRENT STATUS

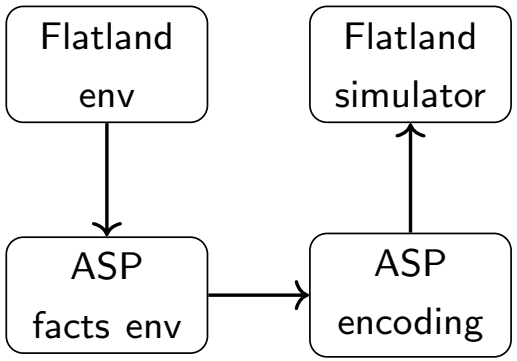
Flaspland

What is Flaspland?

A tool that bridges the gap between Flatland and ASP.

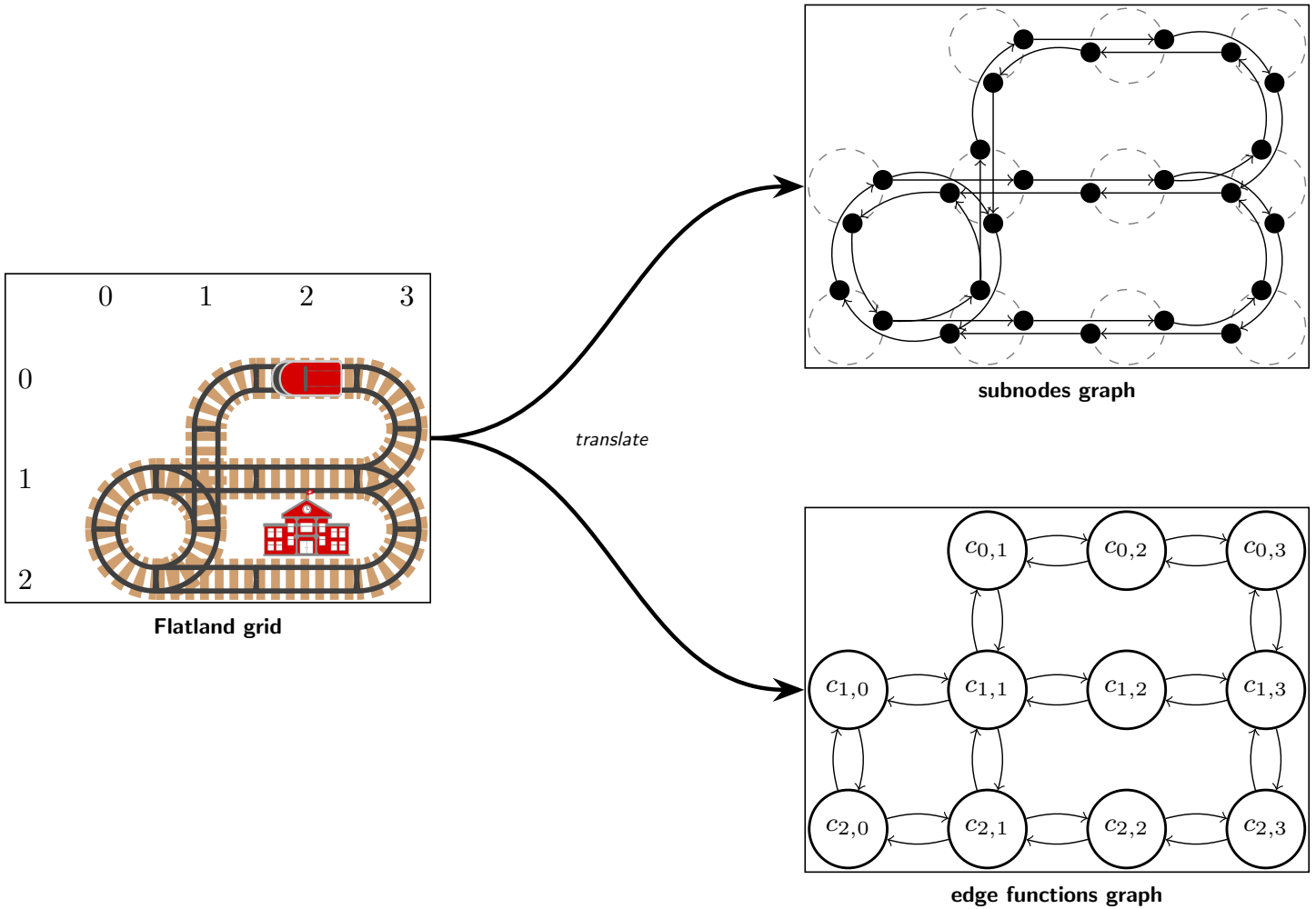
Workflow:

- Generates environments
- Converts environments to ASP facts
- Converts ASP solutions to Flatland actions
- Runs simulation



Environments as graphs

We have a set of representations as alternatives for the Flatland environment.



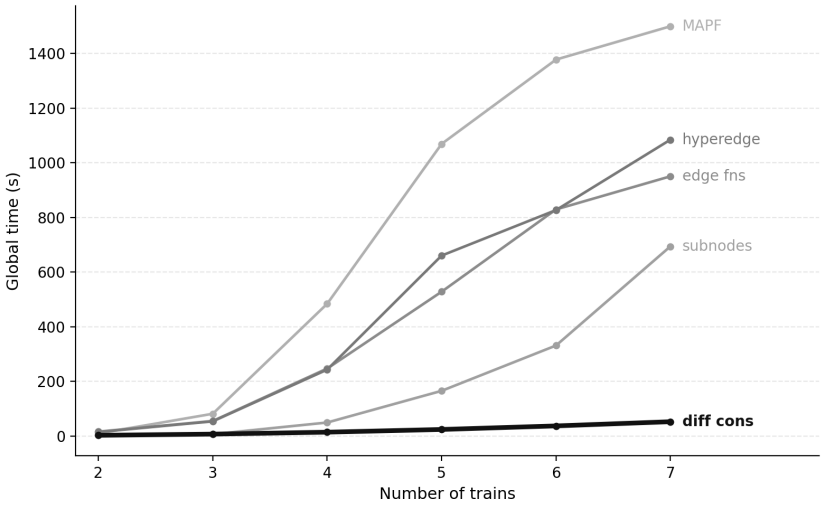
PRELIMINARY RESULTS

Results

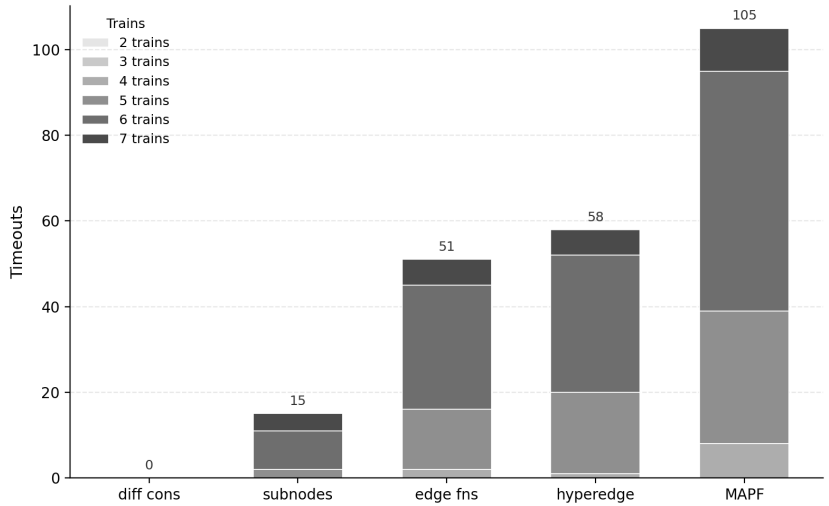
We have run several benchmarks:

- the strongest graph our *subnodes representation*
- advantageous to use difference constraints for pathfinding

Overall time



Number of timeouts

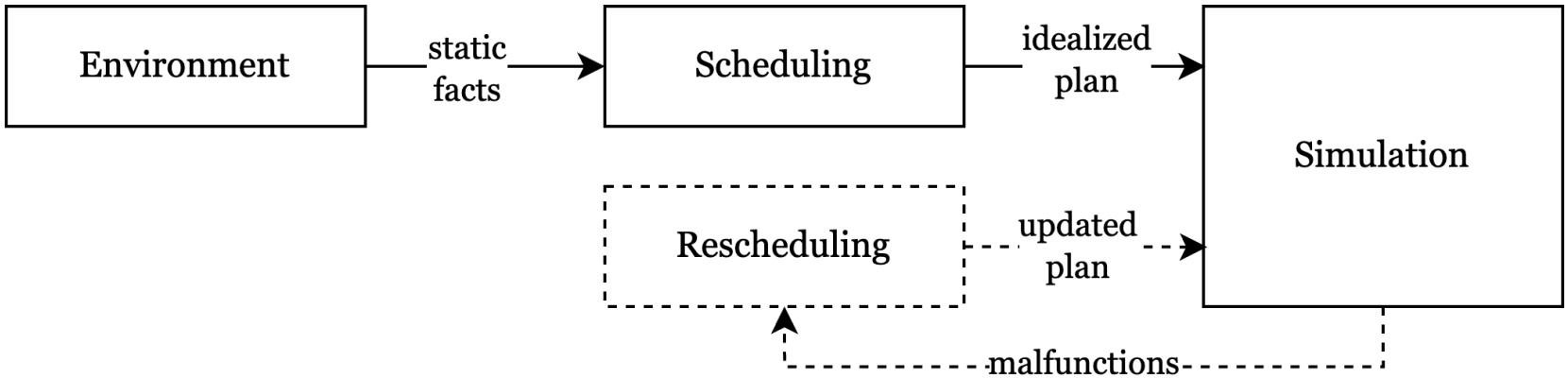


OPEN TOPICS

Open topics

What comes next?

- Continued development of encodings
- Continued exploration of graphs
- Transitioning from VSP to VRSP



QUESTIONS?